Digital Genesis - completed six-hour experiment

Completed normally at the wall-time limit. All 40 populations (four treatments paired across seeds 0-9) advanced exactly **373,250 ticks**. Elapsed time: **21,600.391 seconds** (six hours plus 0.391 s). Start: 06 November 2026 21:26:32 IST. Finish: 07 November 2026 03:26:32 IST. This fresh run completed without a reported interruption; earlier stopped and failed experiments are separate archives and are not combined with these results.

Results

Each treatment began with 30 founders per seed (300 total), with a living cap of 100 per seed, resource fraction 0.35, maturity 100 ticks and lifespan 3,000 ticks. All used the same saved terrain and initial food; SHA-256: 21fd166cece622a20acf3b6e7aa6605d60f4f66a82027bd5c93147b7b874cf76. The requested duty target was 60% of one core.

Treatment	Final survivors (total)	Extinct seeds / 10	Births (total)	Mean observed lifespan	Food efficiency	Mean fitness
Cautious knowledge	20	6	16,990	719.87	2.138	407.22
Original knowledge	10	8	8,554	644.20	2.332	355.13
Ordinary evolution	3	9	7,868	632.93	2.307	350.14
Fixed genomes	13	7	15,972	695.99	2.165	388.04

Lifespan is a mean of each seed's observed ages, including right-censored living individuals. Food efficiency is organism-ticks per food unit consumed, not a biochemical conversion efficiency. Fitness is the original age/food/exploration score averaged across all organisms ever born; it is not an intelligence measure. Survivor totals are endpoint counts, not lifetime survival probabilities. Extinct populations remain extinct; organisms are not revived or replaced automatically.

What the comparisons support

The cautious variant is a hand-designed reserve policy layered on the inherited ambition weight. Its endpoint differences against each control are descriptive, not proof of genetic adaptation. Paired means and sample standard deviations are available in cautious_contrast.csv and knowledge_contrast.csv. Seeds show substantial variation, several populations go extinct, and multiple outcomes were examined. No formal significance claim or general intelligence claim is made.

Compared with the fixed-genome baseline, cautious knowledge has a higher mean fitness and more final survivors, but lower food efficiency; its mean birth and organism-tick advantages vary widely between paired seeds. The original knowledge drive does not establish a consistent advantage over ordinary evolution. These results support further testing of cautious information seeking, not the claim that six hours created smarter organisms. A direct frozen-genome test is needed to separate inherited adaptation from the manually specified policy.

Automatically recorded experience

Treatment	Food-association events	Danger-association events	Exploration-goal completions	Deepest lineage
Cautious knowledge	1,243,249	28,454	136,735	492
Original knowledge	667,253	7,750	70,525	399
Ordinary evolution	0	0	0	488
Fixed genomes	0	0	0	446

These are program-defined, local experience records. Associations can include forgetting and relearning; goals count unique visited locations. Their volume is not evidence of subjective hunger for knowledge, consciousness, or general understanding. Controls without a knowledge module have no such event logger; zero logged events does not imply zero food/danger memory in their existing policy. Lineage depth is parent-to-child depth, not the UI's 400-tick logging-window count. All fruit interactions remained offline. No network service or model was involved.

Reliability and next experiments

The retry-protected CSV publication and periodic exact-state snapshots worked through this run; final manifest status is complete. The latest periodic snapshot can precede the terminal tick by less than 1,000 ticks. Final CSVs and journals were flushed and closed. The desktop windows may remain open, but the experiment has stopped at its configured limit. No new experiment has been started.

Recommended next-run changes, kept separate from this completed experiment:

1. Evaluate frozen early and late evolved genomes on held-out terrain and seeds under identical policies, with predefined survival/reproduction endpoints.

2. Add a fixed-genome cautious-drive control to isolate the manually coded energy reserve rule from mutation and inherited behavioural change.

3. Log actual energy intake/waste and action frequencies; test scarcity and memory ablations to explain extinction and exploration costs. Predefine the primary outcome and increase independent seed counts before drawing strong conclusions.

Files: comparison.csv (40 paired cases), treatment_summary.csv (four aggregate rows), cautious_contrast.csv, knowledge_contrast.csv, history.csv, experiment.json, per-seed ancestry/birth/learning journals and checkpoint.pkl. Earlier artifacts and the first-ancestor memorial remain preserved. Overnight monitoring is now finished and its scheduled follow-up will be removed.